

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 222 — Past Questions & Materials
by Renance

Course: EMT 222
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — EMT 222 Methods in environmental analysis 1 2022/2023

Level 200 • Second Semester • PastQuestion

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DEPARTMENT OF EXVIRONME} TAL MANAGEMENT AND TOXICOLOGY

Lay 4 -% I

SECOND SEMESTER EXAMINATION FOR 2022/2023 ACADEMIC SESSION

DEGREE MARTINA TION

COURSE TITTLE: Methods in Envirenmental Analvsis |

COURSE CODE: EMT 222 3 UNITS) TIME: 2 hours 30 min

INSTRUCTIONS: Answer four questions, at least one from each section

SECTION A

1. (a) Highlight four uses/applications of titrimetric analysis (4 marks)

(b) Explain the following (4 marks) |

(i) Equivalent point

Gi) standard solution

(c) What are two essential precautions when using an analytt

cal balance? (2 mins)

used in titration proces

eae eat 7 et ane Letfou- indicators eormuan)

toy Can you identify anc describe four tpes of titrations comms:

environmental analysis? (marks)

(=, Could vou provide a detailed & =e qation of one of the titrations mentioned earlier

thet are used 11 environmental anak 2:

ic ~ ou list four labora ses commonly utilised ror {irimieiric analysis

ang provide two schemauc

iselble liquids with a relevant :

3. (ay amprehensively explain how wi separate Two Hm

example (5 marks)

'b) Give three example

(c)Name and sketch a diagran of cre \

immiscible liquids (2 marks)

s of inimiscibie liquid solutions (2 starks)

laboratory tool used «n separating Wo

SECTION B :

icsolved oxygen (DO) ug/L) ie sample when 200
f0.0275N sodium thiosulphate titrant? 6 marks)

4. (a) What is the concentration of 4
gen (DO) to the aquatic environment 6

. mL of the sample requires 8.6 mL o

(b)State the significance of dissolved oxy
n"2 (DO)

marks) - . aus Volt red oNVeET

5. (a) Could you please clearly and concisely explain the term 'dissoly ed ONY?!
marks)

Page I of 2

(b) Could you provide a detailed explanation of the steps involved in using an
analytical balance for Precise measurements? (4 ma rks).

(c) What factors might incr
ease dissolved oxypen levels in @ fish pond? (3ma rks}

SECTION GC

& (a) Please list four coy

THON reapents fiz Corer)
merk)

ib) What is the Slanificance Ofalhalinit in the environ
a "A

Our Surroundings? Qmarks)

(¢) Can sou explain the difference betscen the terms °

(d} What does jt signify wher the PH Of soff com
titrant' and 'titrar 3°? Qmarks}

5.9? (2marks) .

ited with diese} js measured at

7 (a) What si¢ps would you tak

s¢ 10 collect w

laboratory analysis of physicochemical! pa

sample's pH js not interfered with? (2

aler samples from Agberoh River for
rameters like pH. while ens

uring that the

(b) As an environmental Scientist. could vou provide a detailed Procedure for

analysing the pH of a Water sample in the laboratory using a pH meter? {3 marks)

(c) Why is it important to calibrate a pH meter?

What does Calibration involve? 2 "Ain

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er hefnve vsing it tn anehoa acorre

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Sasnieiaiabinnes" canter

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Paper 2 of 3 — EMT 222 Environmental method of analysis 2023/2024

Level 200 • Second Semester • PastQuestion

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COLLEGE OF SCIENCE

MENT AND TOXICOLOGY

DEPARTMENT OF ENVIRONMENTAL MANAGE

2023/2024 ACADEMIC SESSION SECOND SEMESTER

EMT 222: Environmental method of analysis II (2 Units)

Instruction: Answer 4 questions, at least one from each section 2 hours

1. (a) Why sampling and analysis are an integral part of data quality, between sampling and analysis, which one often generates more errors? Why?

(b) A new method is being developed to analyse a pesticide in soils. A spiked sample with a known concentration of 17,00 mg/kg was measured five times using the new method (data: 15.3, 17.1, 16.7, 15.5, 17.3 mg/kg, $STD_{new} = 0.923$) and the established EPA method (data: 15.4, 15.9, 16.7, 16.1, 16.2 mg/kg, $STD_{EPA} = 0.472$), Determine the following parameters for both technique.

(i) Is 15.3 mg/kg of the new method data an outlier? if 4h F t

(i) Calculate Coefficient of variation

(iit) Relative Standard Deviation

(iv) Variance

(v) Range

(vi) Are the two methods significantly different at 95% confidence level?

(c) Calculate the relative error for both methods. Which one is more accurate and why?

2. (a) Outline four objectives of environmental sampling and analysis

(b) Three laboratories were given a standard reference material (SRM) and were asked to determine its lead concentration in (mg/kg) using the USEPA standard method. Their results were as follows

Laboratory A - 45.2, 47.3, 51.2, 50.4, 52.2, 48.7 ($STD = 2.62$)

Laboratory B - 49.4, 50.3, 51.6, 52.1, 50.9, 51.7 ($STD = 1.01$)

Laboratory C - 48.3, 47.7, 51.6, 55.7, 52.4, 51.5 ($STD = 2.92$)

(i) For laboratory C, is 55.7 mg/kg a possible outlier?

(ii) Calculate the RSD for each laboratory: which laboratory is more precise

the percentage

vs ; if 49.5 mg/kg, calculate .

(iii) If the known concentration of lead in SRM is 49 ; OE caatt (Note: without \

recovery for each laboratory. Which laboratory |

considering the deletion of outlier if any)? . idence level?

(iv) Are laboratory A & B results significantly different at 95% confi

the primary means of

(c) The table below has a list of quality control (QC) parameters ate ae procedure. Explain

general QC procedure. Write out the correct letter each corresponding

| QC parameter Place a letter here | QC rocedure ae

1. Accuracy a, Analysis of blanks ;

2. Precision b. Analysis of matrix S ikes

x F3 Ss

3. Cross-contamination c. Analysis of replicate sam le

4. Extraction efficiency d. Analysis of reference materials or

n concentration

samples of know

TABLE 7-5

Critical Values for the Rejection Quotient, Q"

Ocorr (Reject if O > Qais)

_| Number of © ;

~ Observations ———«90-%e Confidence 95% Confidence 99% Confidence

3 0.941 0.970 0.994

4 0.765 0.829 0.926

5 0.642 0.710 0.821

6 0.560 0.625 0.740

7 0.507 0.568 0,680

8 0.468 0.526 0,634

9 0.437 0.493 0.598

10 0.412 0.466 0.568

Section B

1. Discuss the importance of measuring pH in environmental studies, focusing on how pH affects water quality, aquatic ecosystems, and the behaviour of pollutants in the environment.

2. Explain how alkalinity is related to a water sample's ability to resist pH changes, particularly how it acts as a buffer against acidification and what factors contribute to = acid-neutralizing capacity.

3 Describe why measuring acidity is critical in environmental monitoring, focusing on how elevated acidity levels can affect aquatic life, corrode infrastructure, and indicate the presence of pollutants or environmental degradation.

Section C

- i a. State five precautions to take when using an analytical balance.
 - c. Describe the process of calibrating a burette and pipette, highlighting the importance of accuracy and precision in each step.
- 2, a. List two indicators commonly used in titration, and explain their relevance in determining
- b, Differentiate between hard water and soft water, and provide examples of compounds responsible for hardness and softness.
 - c, Outline the step-by-step procedure for determining total hardness in a water sample, including any necessary reagents and equipment.

Paper 3 of 3 — EMT 222

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